Chapter 27

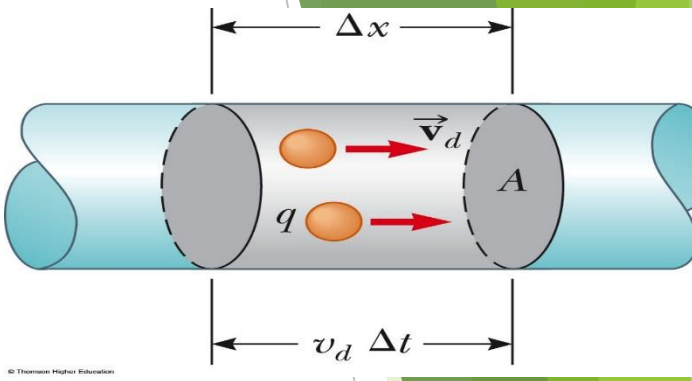
Current And Resistance

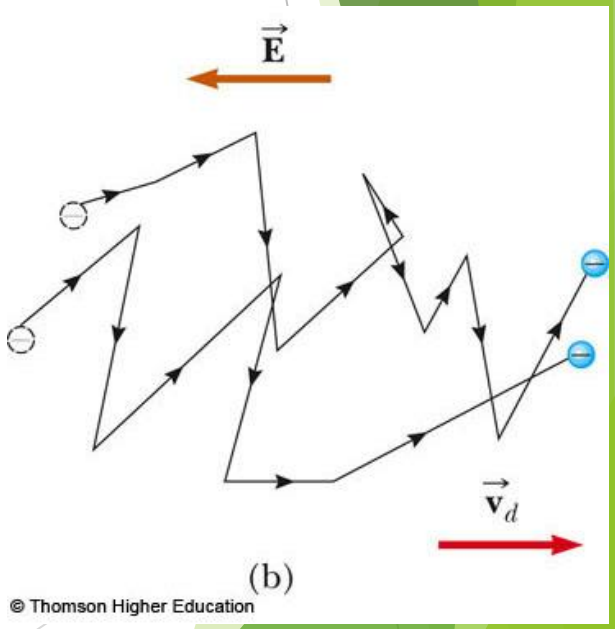
Sec 27.1: Electric Current

- **Electric current** is the rate of flow of charge through some region of space
- The SI unit of current is the **ampere** (A)
  - 1 A = 1 C / s
- The symbol for electric current is I
- Average Electric Current : $$I_{avg} = \frac{\Delta Q}{\Delta t}$$
- Instantaneous Electric Current: $$I \equiv \frac{dQ}{dt}$$
- The charges passing through the area could be positive or negative or both
- It is conventional to assign to the current the same direction as the flow of positive charges

Direction of Current

- The direction of current flow is opposite the direction of the flow of electrons
- It is common to refer to any moving charge as a *charge carrier*
- $I_{avg} = nqv_dA$
- The zigzag black lines represent the motion of a charge carrier in a conductor
  - The net drift speed is small
- The sharp changes in direction are due to collisions
- The net motion of electrons is opposite the direction of the electric field





Motion of Charge Carriers, final

- The electric field set up by the battery exerts forces on the conduction electrons in the wire
- These forces cause the electrons to move in the wire and create a current
- The battery does not supply the electrons, it only establishes the electric field

Quick Quiz 27.1 Consider positive and negative charges moving horizontally through the four regions shown in Figure 27.4. Rank the current in these four regions from highest to lowest.

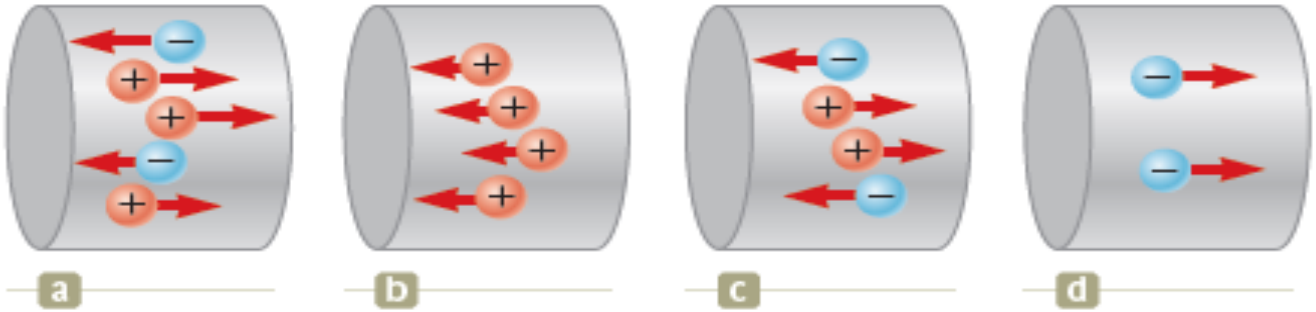


Figure 27.4 (Quick Quiz 27.1) Charges move through four regions.

Sec 27.2: Resistance

- J is the **current density** of a conductor
- It is defined as the current per unit area
  - $J = I / A = nqv_d$
  - This expression is valid only if the current density is uniform and A is perpendicular to the direction of the current
- J has SI units of A/m^2
- The current density is in the direction of the positive charge carriers
- For some materials, the current density is directly proportional to the field
- The constant of proportionality, σ, is called the **conductivity** of the conductor
- **Ohm's law** states that for many materials, the ratio of the current density to the electric field is a constant σ that is independent of the electric field producing the current

- Most metals obey Ohm's law
- Mathematically, $J = \sigma E$
- Materials that obey Ohm's law are said to be *ohmic*
- Materials that do not obey Ohm's law are said to be *nonohmic*
- In a conductor, the voltage applied across the ends of the conductor is proportional to the current through the conductor

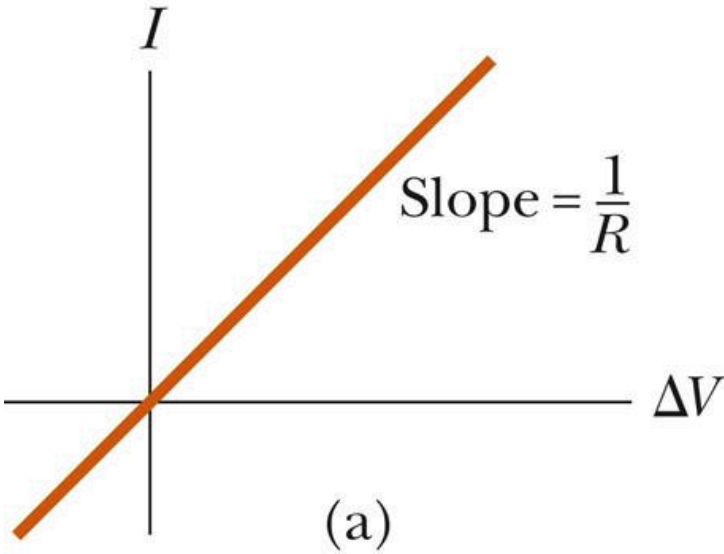
$$R \equiv \frac{\Delta V}{I}$$

- The constant of proportionality is called the **resistance** of the conductor
- SI units of resistance are ohms (Ω), 1 Ω = 1 V / A
- Resistance in a circuit arises due to collisions between the electrons carrying the current with the fixed atoms inside the conductor

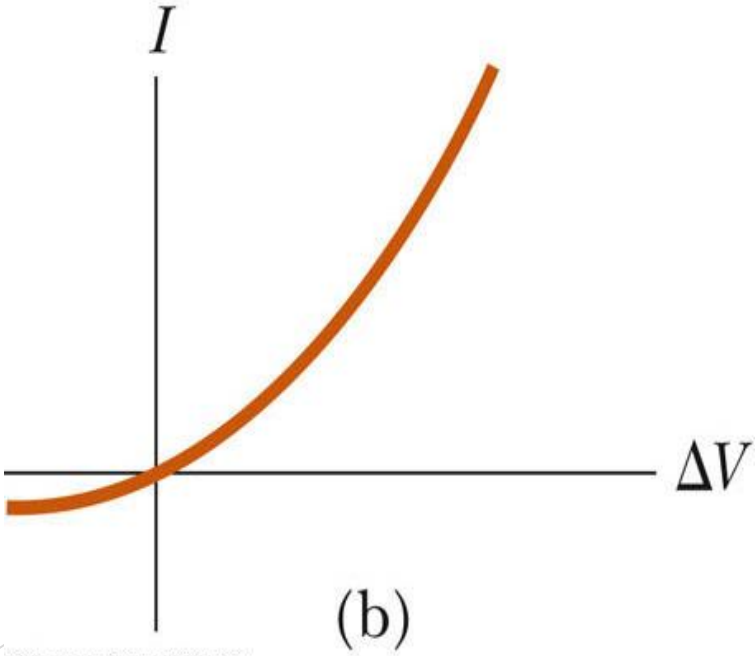
Resistivity

- The inverse of the conductivity is the **resistivity**:
  - $\rho = 1 / \sigma$
- Resistivity has SI units of ohm-meters ($\Omega \cdot$ m)
- Resistance is also related to resistivity: $$R = \rho \frac{\ell}{A}$$
- Every ohmic material has a characteristic resistivity that depends on the properties of the material and on temperature
  - Resistivity is a property of substances
- The resistance of a material depends on its geometry and its resistivity
  - Resistance is a property of an object
- An ideal conductor would have zero resistivity
- An ideal insulator would have infinite resistivity

- An ohmic device
- The resistance is constant over a wide range of voltages
- The relationship between current and voltage is linear
- The slope is related to the resistance



- Nonohmic materials are those whose resistance changes with voltage or current
- The current-voltage relationship is nonlinear
- A junction diode is a common example of a nonohmic device



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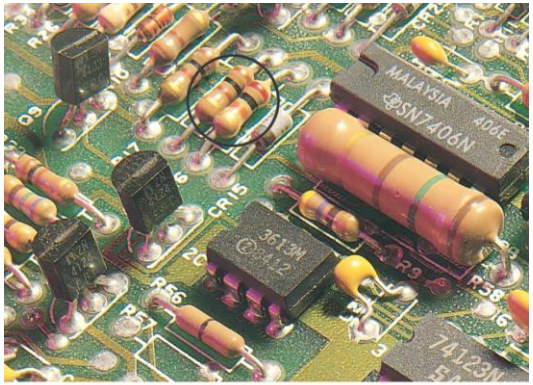
Resistor Values



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- Values of resistors are commonly marked by colored bands

TABLE 27.1

Color Coding for Resistors

Color	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%
Colorless			20%

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Resistivity Values

TABLE 27.2

Resistivities and Temperature Coefficients of Resistivity for Various Materials

Material	Resistivity[a] ($\Omega \cdot m$)	Temperature Coefficient[b] $\alpha[(°C)^{-1}]$
Silver	1.59×10^{-8}	3.8×10^{-3}
Copper	1.7×10^{-8}	3.9×10^{-3}
Gold	2.44×10^{-8}	3.4×10^{-3}
Aluminum	2.82×10^{-8}	3.9×10^{-3}
Tungsten	5.6×10^{-8}	4.5×10^{-3}
Iron	10×10^{-8}	5.0×10^{-3}
Platinum	11×10^{-8}	3.92×10^{-3}
Lead	22×10^{-8}	3.9×10^{-3}
Nichrome[c]	1.50×10^{-6}	0.4×10^{-3}
Carbon	3.5×10^{-5}	-0.5×10^{-3}
Germanium	0.46	-48×10^{-3}
Silicon[d]	2.3×10^{3}	-75×10^{-3}
Glass	10^{10} to 10^{14}	
Hard rubber	$\sim 10^{13}$	
Sulfur	10^{15}	
Quartz (fused)	75×10^{16}	

[a] All values at 20°C. All elements in this table are assumed to be free of impurities.

[b] See Section 27.4.

[c] A nickel–chromium alloy commonly used in heating elements.

[d] The resistivity of silicon is very sensitive to purity. The value can be changed by several orders of magnitude when it is doped with other atoms.

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Quick Quiz 27.2 A cylindrical wire has a radius r and length ℓ. If both r and ℓ are doubled, does the resistance of the wire (a) increase, (b) decrease, or (c) remain the same?

Example 27.2 The Resistance of Nichrome Wire

The radius of 22-gauge Nichrome wire is 0.32 mm.

(A) Calculate the resistance per unit length of this wire.

$$\frac{R}{\ell} = \frac{\rho}{A} = \frac{\rho}{\pi r^2} = \frac{1.0 \times 10^{-6}\,\Omega \cdot \mathrm{m}}{\pi(0.32 \times 10^{-3}\,\mathrm{m})^2} = \boxed{3.1\ \Omega/\mathrm{m}}$$

(B) If a potential difference of 10 V is maintained across a 1.0-m length of the Nichrome wire, what is the current in the wire?

$$I = \frac{\Delta V}{R} = \frac{\Delta V}{(R/\ell)\ell} = \frac{10\ \mathrm{V}}{(3.1\ \Omega/\mathrm{m})(1.0\ \mathrm{m})} = \boxed{3.2\ \mathrm{A}}$$

Sec 27.4: Resistance and Temperature

- Over a limited temperature range, the resistivity of a conductor varies approximately linearly with the temperature

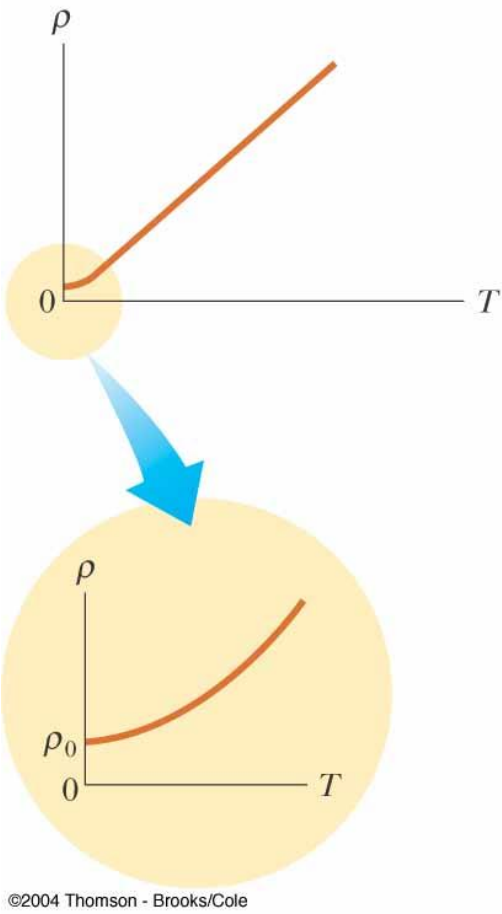
$$\rho = \rho_o[1 + \alpha(T - T_o)]$$

- ρ_o is the resistivity at some reference temperature T_o
- T_o is usually taken to be 20° C
  - α is the **temperature coefficient of resistivity**
    - SI units of α are $^oC^{-1}$
- Since the resistance of a conductor with a uniform cross-sectional area is proportional to the resistivity, you can find the effect of temperature on resistance

$$R = R_o[1 + \alpha(T - T_o)]$$

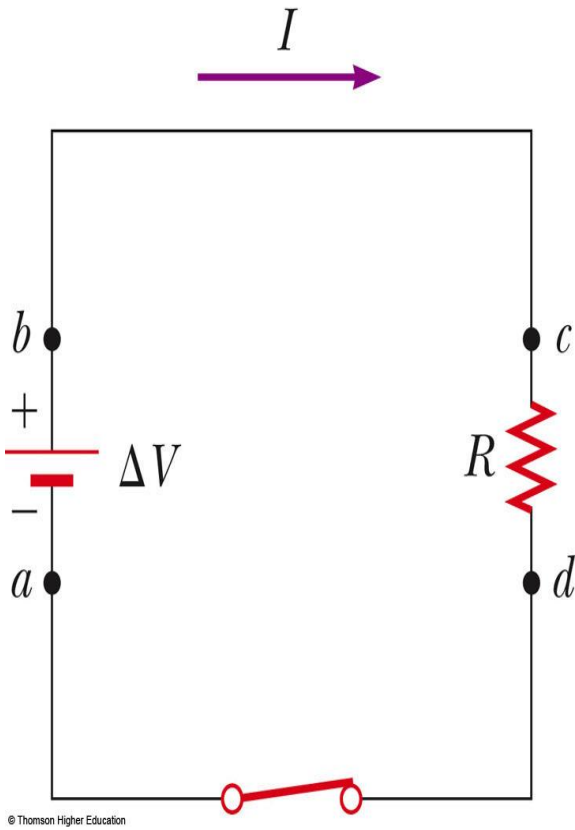
Resistivity and Temperature, Graphical View

- For some metals, the resistivity is nearly proportional to the temperature
- A nonlinear region always exists at very low temperatures
- The resistivity usually reaches some finite value as the temperature approaches absolute zero
- **Semiconductors** are materials that exhibit a decrease in resistivity with an increase in temperature
- α is negative
- There is an increase in the density of charge carriers at higher temperatures



Sec 27.6: Electrical Power

- As a charge moves from a to b, the electric potential energy of the system increases by $Q\Delta V$
  - The chemical energy in the battery must decrease by this same amount
- As the charge moves through the resistor (c to d), the system loses this electric potential energy during collisions of the electrons with the atoms of the resistor
- This energy is transformed into internal energy in the resistor
  - Corresponds to the increased vibrational motion of the atoms in the resistor

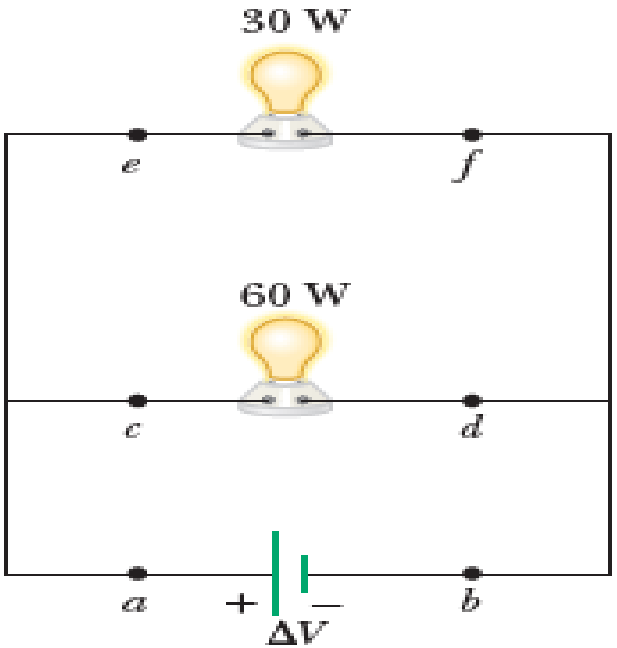


- The resistor is normally in contact with the air, so its increased temperature will result in a transfer of energy by heat into the air, also emits thermal radiation
- After some time interval, the resistor reaches a constant temperature
- The **power** is the rate at which the energy is delivered to the resistor
- The power is given by the equation: $\wp = I \Delta V$
- Applying Ohm's Law, alternative expressions can be found:

$$\wp = I \Delta V = I^2 R = \frac{(\Delta V)^2}{R}$$

- Units: I is in A, R is in Ω, V is in V, and $\wp$ is in W

Quick Quiz 27.5 For the two lightbulbs shown in Figure 27.13, rank the current values at points *a* through *f* from greatest to least.



Example 27.4 Power in an Electric Heater

An electric heater is constructed by applying a potential difference of 120 V across a Nichrome wire that has a total resistance of 8.00 Ω. Find the current carried by the wire and the power rating of the heater.

$$I = \frac{\Delta V}{R} = \frac{120\ \text{V}}{8.00\ \Omega} = \boxed{15.0\ \text{A}}$$

$$P = I^2R = (15.0\ \text{A})^2(8.00\ \Omega) = 1.80 \times 10^3\ \text{W} = \boxed{1.80\ \text{kW}}$$